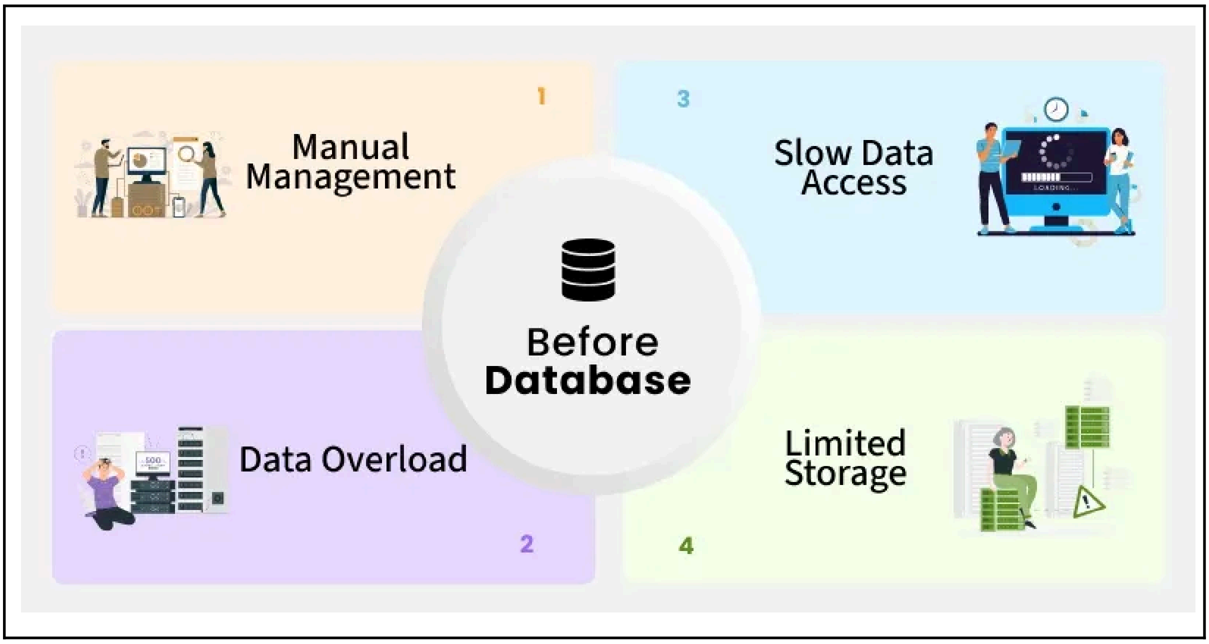
Databases

Databases are systems that allow users to store and organize data. They are technically useful when dealing with large amounts of data.



Who uses databases?

Databases have a wide variety of users:

- Analysts:
    - Marketing
    - Business
    - Sales
- Technical:
    - Data Engineers
    - Data Scientist
    - Software Engineers
    - Web Developers

Basically anyone needing to deal with data

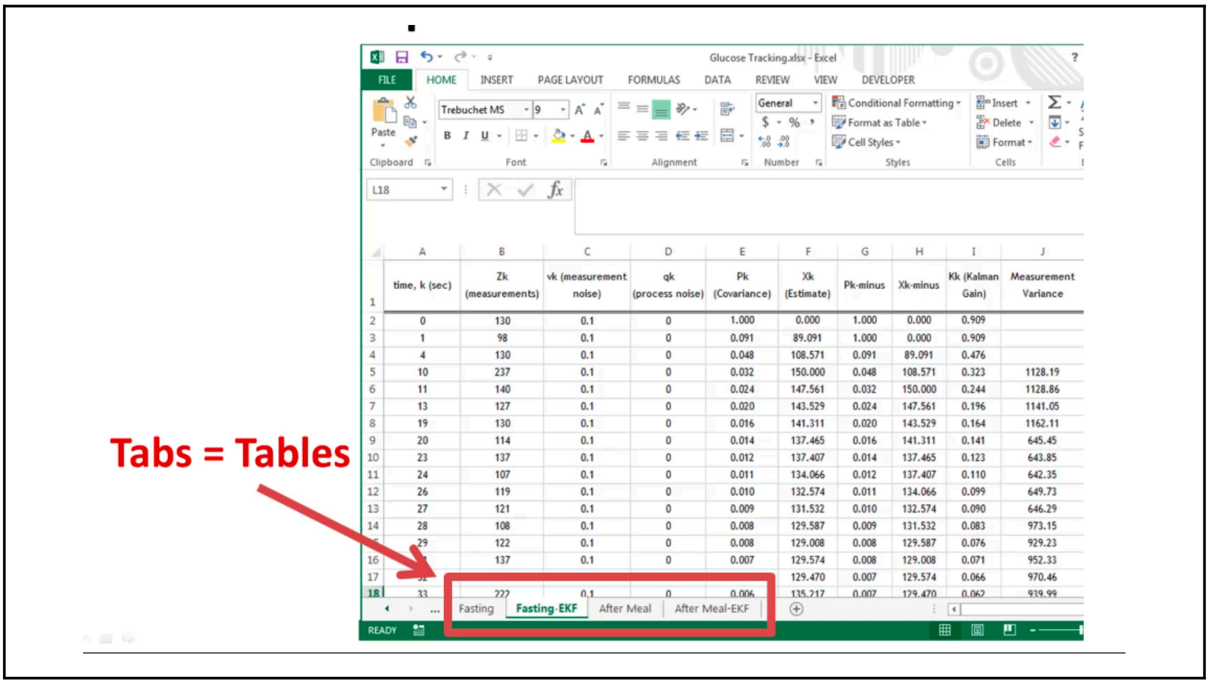
Spreadsheets (Excel) Vs Databases

Most users have some familiarity with spreadsheet software (such as Excel). The use cases of spreadsheets vary from use cases of databases

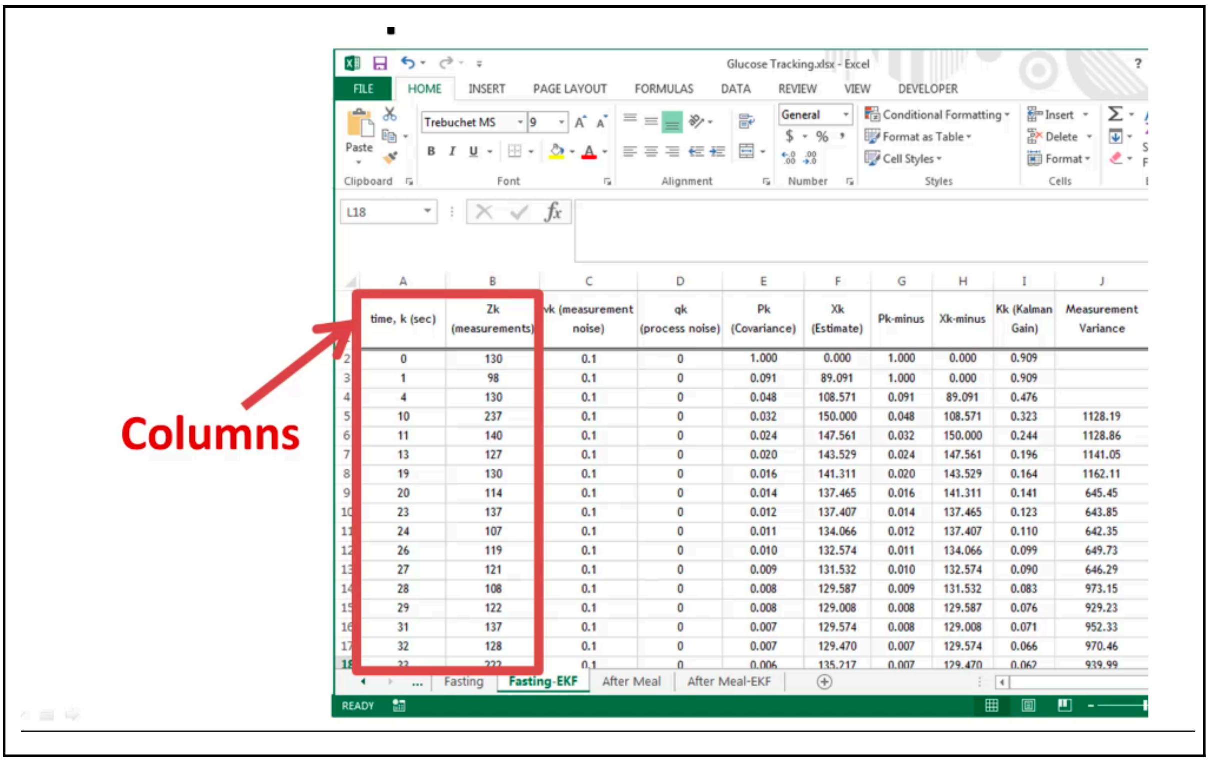
Spreadsheets	Databases
Really good for one time analysis	Data integrity
Quickly need to chart something out	Can handle massive amounts of data
Reasonable dataset size	Can handle massive amounts of data
Ability for untrained people to work with data	Quickly combine different datasets
	Automate steps for re-use
	Can support data for websites and applications

Most users have some familiarity with spreadsheet software (such as excel), so they can easily leverage that knowledge from spreadsheets to understand databases.

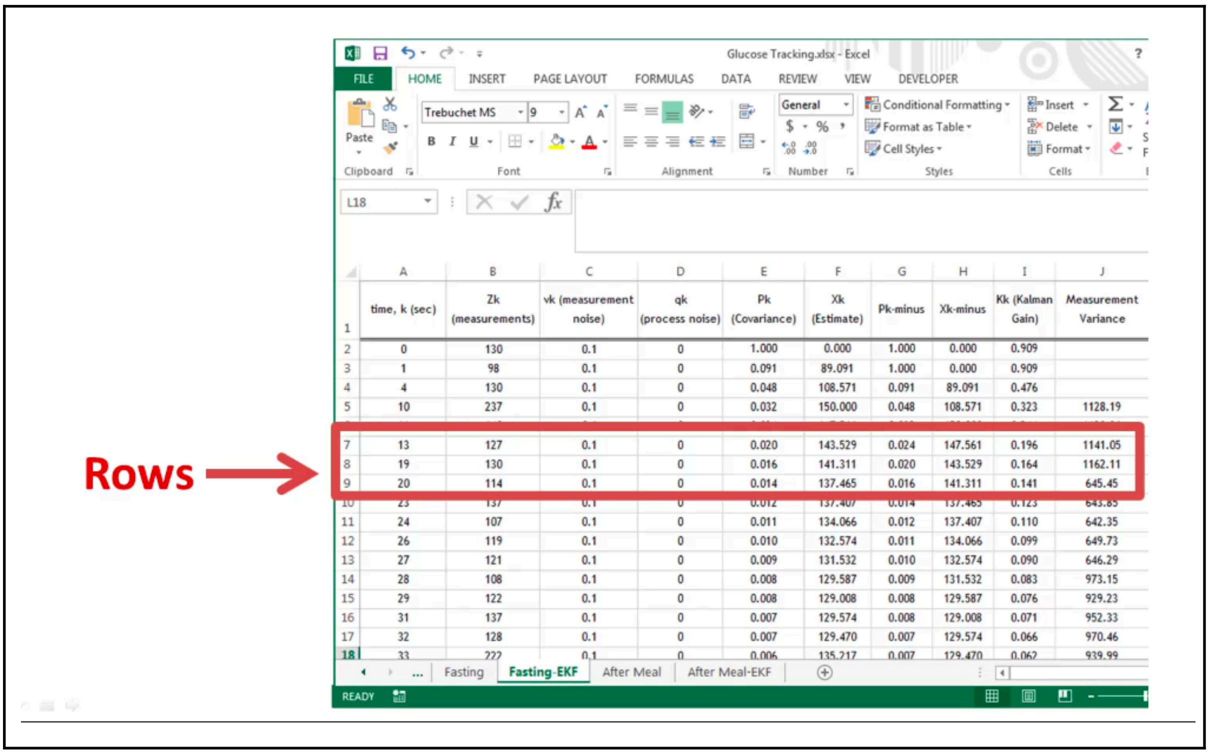
Excel Tabs -> SQL Table



Excel Columns -> SQL Columns



Excel Rows -> SQL Records



Database Platform Options

Main Ones (Not all of them) :

PostgreSQL	**Free** (Open Source) Widely used on internet Multi platform
MySQL MariaSQL	**Free** (Open Source) Widely used on internet Multi platform.
MS SQL Server Express	**Free**, but with some limitations Compatible with SQL Server Windows only
Microsoft Access	Cost (-) Not easy to use just SQL (-)
SQLite	Free (Open Source) Mainly command line (-)